CREDIT CARD FRAUD
PATTERN ANALYSIS

A Detection-Led Approach

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Dataset: Worldline/ULB Credit Card Fraud Detection (Kaggle) | 284,807 transactions

Project Overview

284,807	492	0.17%	£60.1K
Total Transactions	Fraud Cases	Fraud Rate	Total Fraud Value

Project Objectives

1. Identify fraud patterns using SQL and Python on a real-world anonymised dataset
2. Build detection rules grounded in fraud domain knowledge
3. Develop and compare three distinct scoring models with evidence-based thresholds
4. Exceed the UK Finance 2024 benchmark of 67% recall across all final models

Infrastructure

BigQuery	fraud-analytics-project1
Dataset	fraud_analysis
Table	credit_card_transactions
Cloud Storage	europe-west2 (London)
Python	Google Colab
GitHub	fishy08/credit-card-fraud-analysis

Dataset and Key Assumptions

Time	V1–V28	Amount	Class
Seconds elapsed since first transaction	PCA-anonymised features (28 components)	Transaction value in Euros	0 = Genuine 1 = Fraud

⚠ Important Assumption — CNP Transaction Type

The dataset does not explicitly confirm Card Not Present (CNP) transactions. The CNP assumption is informed by: (1) zero-amount transactions consistent with digital wallet token registration; (2) small-amount velocity bursts consistent with BIN attacks; (3) absence of location and merchant data consistent with PCA anonymisation of CNP metadata. All CNP references are labelled as informed assumptions throughout the analysis.

Phase 1	Phase 2	Phase 3	Phase 4
SQL Quality Checks	Python EDA	Detection Models	Power BI Dashboard

Phase 1 — SQL Data Ingestion and Quality Checks

Query	Purpose	Key Finding
P1_Q01	Class distribution — fraud vs genuine count	492 fraud (0.17%) | 284,314 genuine (99.83%)
P1_Q02	Null check across all 31 columns	Zero nulls — dataset complete
P1_Q03	Duplicate Time + Amount + Class combinations	Duplicates confirmed — three legitimate explanations identified
P1_Q04	Duplicate fraud transaction isolation	100% fraud rate on duplicate patterns — BIN attack confirmed
P1_Q05	Amount distribution by class — min, max, avg	Fraud avg £122 vs genuine £88 — threshold avoidance confirmed

Zero-Amount Transactions

£0 transactions are NOT data errors. They represent card verification checks, digital wallet token registrations, and CNP pre-authorisations.

Fraud has a 9x higher zero-amount rate than genuine transactions (5.49% vs 0.63%).

BIN Attack Confirmation

Duplicate fraud patterns — identical Time, Amount, and Class — confirm coordinated BIN attacks where a batch of generated card numbers is tested simultaneously at the same merchant for the same amount.

Threshold Avoidance

Maximum fraud amount £2,125 vs £25,691 legitimate. Fraudsters deliberately self-limit transaction amounts below levels they believe trigger high-value fraud rules — a recognised evasion behaviour.

Phase 2 — Fraud Pattern Analysis

Key Behaviours Confirmed

50.6% of all fraud falls under £10 — card testing confirmed

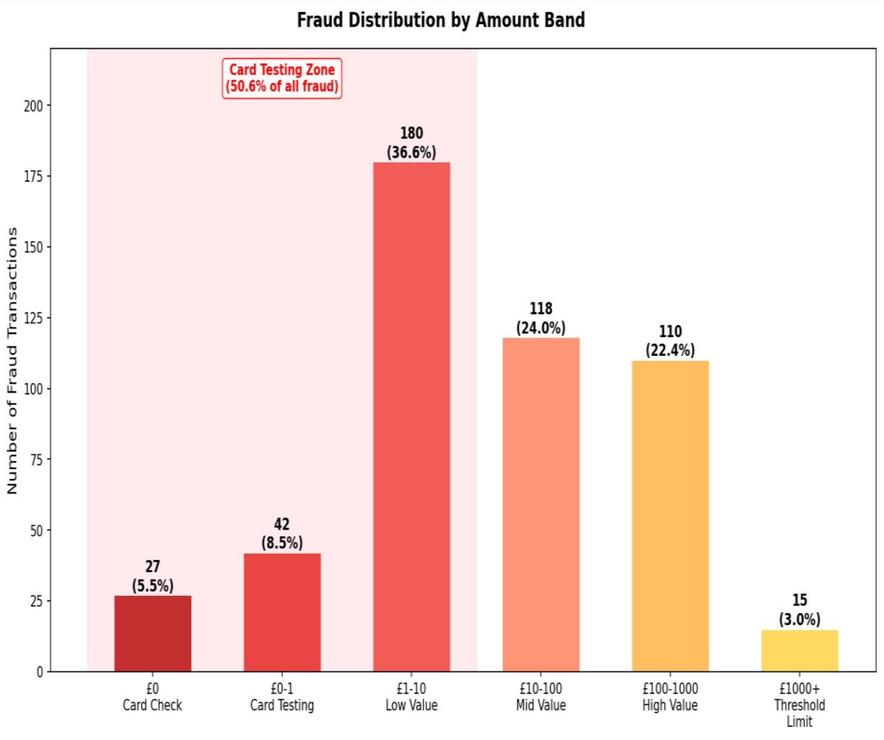
9x higher zero-amount rate in fraud vs genuine transactions

1.70% fraud rate at 2am — 10x the daily average of 0.17%

100% fraud rate on zero-amount + BIN attack velocity bursts

£122 average fraud value vs £88 genuine — threshold avoidance

Fraud by Amount Band



Phase 2 — V Feature Fraud Separation

Feature	Direction	Separation Score	Fraud Rate at Extreme	Rule Threshold
V3	Negative	7.05	36.9% at < -10 | 75.9% at < -20	< -2 (stepped bonus to -20)
V14	Negative	6.98	91.8% at extreme band	< -2
V17	Negative	6.68	100.0% at < -18.28	< -2 (hard flag at -18.28)
V10	Negative	5.69	100.0% at < -14.92	< -2 (hard flag at -14.92)
V9	Negative	—	100.0% at < -7.63	Hard flag at -7.63
V18	Negative	—	100.0% at < -6.59	Hard flag at -6.59
V11	Positive	4.31	86.5% at > +8.66	> +2
V4	Positive	3.87	50.0% at > +7.85	> +2

Near-zero inter-correlation between top V features confirms each adds independently measurable fraud signal. Combining multiple V feature rules in the scoring model is statistically justified — it increases detection confidence without double-counting the same underlying pattern.

Phase 3 — Operational Detection Rules

01 Card Testing

CONDITION

Amount ≤ £10

ACTION

Add fraud score

Card testing is a verification step before fraud attempt

02 Zero Amount token

CONDITION

Amount = £0

ACTION

Add fraud score

Fraud has 9x higher zero-amount rate. These are merchant and digital wallet registrations.

03 Overnight Elevated Risk

CONDITION

Transaction between midnight and 5am

ACTION

Apply risk multiplier — do not block standalone

2am fraud rate 1.70% — 10x the daily average. Standalone overnight transactions are not blocked to avoid customer friction

04 V Feature Extreme Value

CONDITION

V9 < -7.63 | V17 < -18.28 | V10 < -14.92 | V18 < -6.59

ACTION

Instant hard flag — bypass scoring engine entirely

100% fraud rate with zero genuine transactions confirmed across all four bands in the full 284,806 row dataset

Three Detection Model Architecture

Fixed Threshold

13 rules | Max: 22 pts

Amount ≤ £10 (+2), Amount = £0 (+2), Hour 0–5am (+1)

V3, V14, V17, V12, V10, V11, V4 regular rules

V3 stepped extreme: < -10 (+1), < -16 (+2), < -20 (+2)

+ 4 hard flag rules (auto-flag, bypass scoring)

Tiered Variable

44 rules | Max: ~90 pts

28 V features each split into 5 equal bands

Points by fraud rate bracket:

≥75% = full tier max | 50–74% = 60%

25–49% = 40% | 1–24% = 20%

+ Scaled Amount/Time rules (+1, +1, +0.5)

Tiered Proportional

44 rules | Max: ~90 pts

Same 28 V features, same 5 bands per feature

Points = (Fraud Rate% ÷ 100) × Tier Max

Continuously proportional — finer discrimination

between bands with similar fraud rates

+ Scaled Amount/Time rules (+0.5, +0.5, +0.25)

Hard Flag Architecture — 100% Precision Layer

V9 < -7.63

23 fraud | 0 genuine | 100%

V17 < -18.28

43 fraud | 0 genuine | 100%

V10 < -14.92

27 fraud | 0 genuine | 100%

V18 < -6.59

51 fraud | 0 genuine | 100%

69 fraud auto-flagged | 0 genuine | 100% precision

Two-Layer Detection Flow

Transaction arrives

Hard Flag check (V9/V17/V10/V18)

If triggered → Instant auto-flag (100% precision)

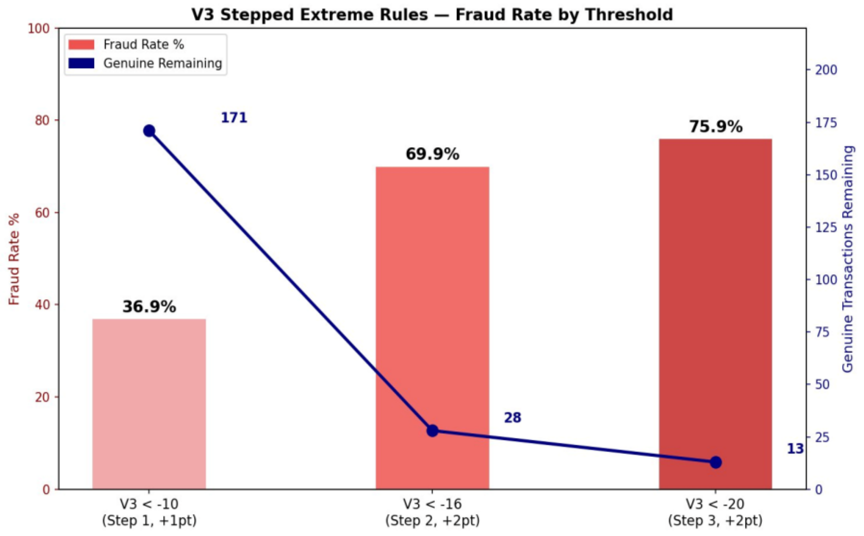
If not → Calculate risk score from 13 rules

If score ≥ T8 (8+ points) → Flag for review

If score < T8 → Pass

Model Enhancement — Systematic Investigation

V3 Stepped Extreme Rules



Step	Condition	Bonus Points	Fraud Rate
1	V3 < -10	+1	36.9% (171 genuine remain)
2	V3 < -16	+2	69.9% (28 genuine remain)
3	V3 < -20	+2	75.9% (13 genuine remain)

Hard Flag Search — Three Directions

Direction 1 **V Feature Combinations**

Tested 11 standard + 10 deep threshold pairs. Best result: V17 & V12 at 80.8% fraud rate — genuine transactions persist at all levels. No new hard flags.

Direction 2 **Amount + V Feature Combos**

Amount = £0 & V10 < -2 reached 88.5% fraud rate (3 genuine remaining). Tested as +2 bonus rule — rejected as it reduced F1 from 0.722 to 0.720.

Direction 3 **All 28 V Features Scan**

Systematic scan at 1st, 2nd, 5th percentile extremes across all 28 features. No additional zero-genuine zones found.

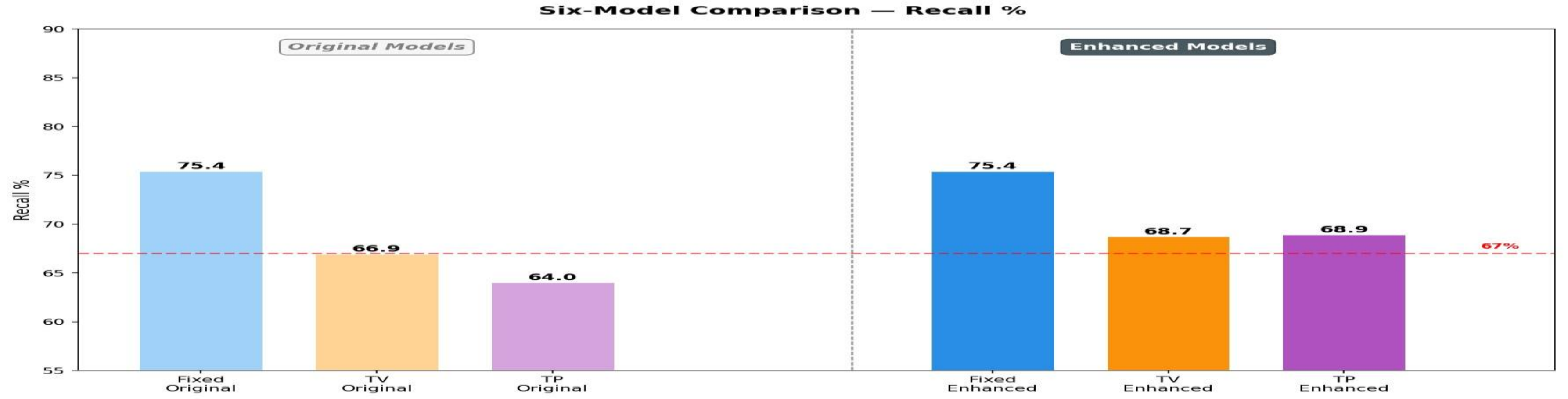
The four existing hard flags are the only zero-genuine extreme zones in this dataset

Final Results — Enhanced Models vs UK Finance 67% Benchmark

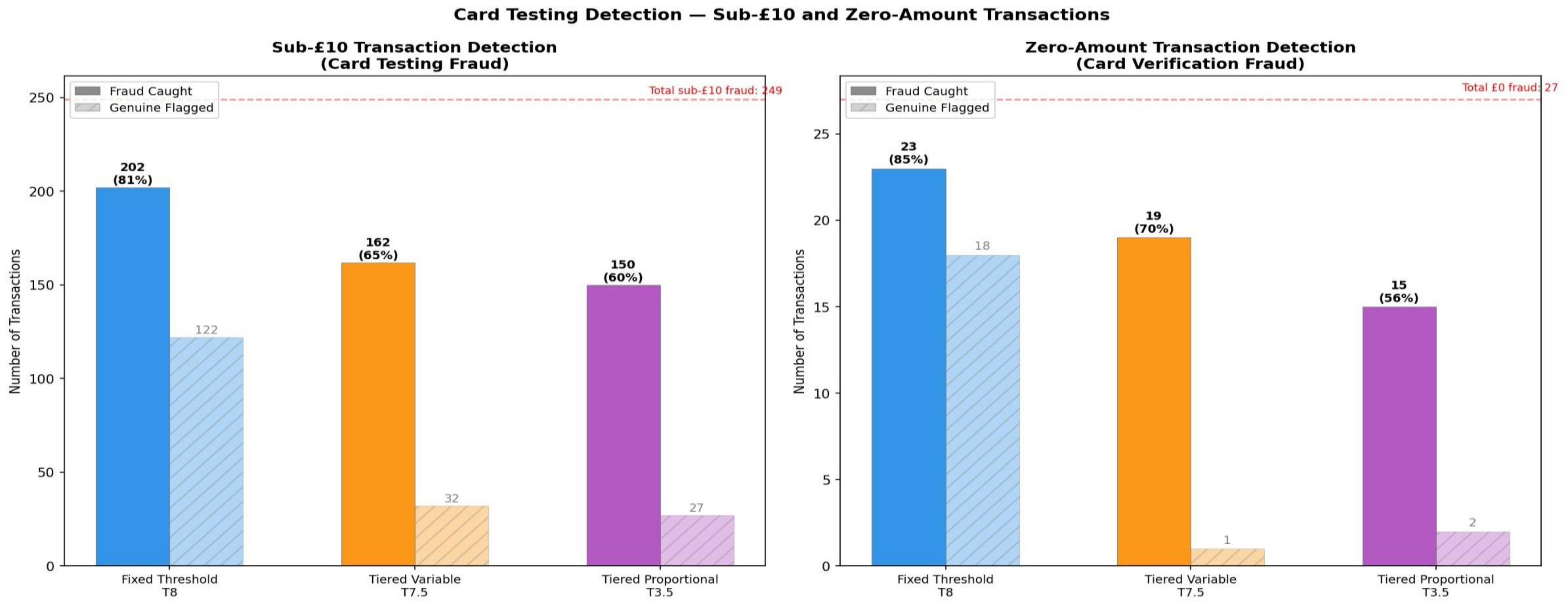
Model	Threshold	Fraud Caught	Flagged	Recall %	Precision %	F1	vs 67%
Fixed Threshold (Original)	T8	371	548	75.4%	67.7%	0.713	✅
Tiered Variable (Original)	T7	329	424	66.9%	77.6%	0.718	❌
Tiered Proportional (Original)	T4	315	406	64.0%	77.6%	0.702	❌
Fixed Threshold ★	T8	371	536	75.4%	69.2%	0.722	✅
Tiered Variable ★	T7.5	338	454	68.7%	74.4%	0.715	✅
Tiered Proportional ★	T3.5	339	498	68.9%	68.1%	0.685	✅

★ Enhanced models

Recall Comparison — Original vs Enhanced



Card Testing Detection — Sub-£10 and Zero-Amount Analysis



Fixed Threshold

Highest catch rate (81.1% sub-£10, 85% £0) — but 122 genuine sub-£10 transactions falsely flagged. Best where maximising detection is the priority.

Tiered Variable

65.1% sub-£10 and 70.4% £0 fraud caught through V features alone. Only 32 and 1 genuine falsely flagged respectively — 4x better false positive rate.

Tiered Proportional

60.2% sub-£10 detection with only 27 genuine false positives. Most conservative — best where analyst capacity is limited.

Conclusions and Recommendation

Recommended Model: Fixed Threshold

Highest recall (75.4%), best F1 (0.722), two-layer hard flag architecture. Where analyst capacity is limited, Tiered Variable offers the best precision-recall balance (68.7% recall, 74.4% precision, only 454 flagged transactions).

Key Conclusions

✓ All three enhanced models exceed UK Finance 2024 benchmark
✓ Hard flags provide 100% precision — 69 fraud auto-flagged with zero false positives
✓ V3 stepped extreme rules add precision without clean zero-genuine zone requirement
✓ V feature combination rules do not produce additional hard flag candidates
✓ Scaled Amount/Time rules push Tiered Proportional recall from 64% to 68.9%

Limitations

— CNP transaction type is an informed assumption — not confirmed
— Location anomaly (primary CNP signal) is unmeasurable in PCA data
— Dataset covers ~2 days — no seasonal analysis possible
— Rule-based models require periodic manual review

Further Development

→ Apply ML classifiers (Random Forest, XGBoost) and compare
→ Add velocity features — transactions per card per hour
→ Build real-time scoring pipeline in BigQuery Streaming

Thank You

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GitHub	github.com/fishy08/credit-card-fraud-analysis
Dataset	Worldline/ULB Credit Card Fraud Detection — Kaggle
Tools	BigQuery | Google Colab | Power BI | Python